International Islamic University Chittagong

ShikkaLink

Providing Learning Excellence

Software Requirements Specification

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1 Introduction

Private tutoring plays a significant role in supporting students' academic progress in Bangladesh. However, finding and managing tutors remains largely informal through social media, personal recommendations, or coaching centers. These methods often result in lack of verification, scheduling difficulties, and trust issues.
This Software Requirements Specification (SRS) document describes the requirements for ShikkaLink, a centralized web-based platform that connects students/guardians with qualified tutors in a secure, organized, and efficient virtual environment.

1.1 Purpose

The purpose of this SRS is to provide a clear, complete, and unambiguous description of the functional and non-functional requirements of the ShikkaLink platform. It serves as the primary reference for the development team, supervisor, and stakeholders.

1.2 Scope

ShikkaLink shall enable students/guardians to search and book qualified tutors, allow tutors to manage profiles and sessions, and provide administrators with tools to maintain platform quality and trust. The system shall support one-on-one online tutoring sessions through external meeting links.
The platform shall be a web-based application with three primary user roles: Student, Tutor, and Administrator.

Out of scope for the initial version:

- Separate guardian-specific features (combined student–guardian interface will be used).
- Offline tutoring sessions.
- Advanced payment processing or native mobile applications.

1.3 Overview

This document is organized as follows:

- Section 2 presents User Requirements Definition.
- Section 3 describes the high-level System Architecture.
- Section 4 provides the detailed System Requirements Specification.
- Section 5 presents System Models.
- Section 6 discusses System Evolution.
- Appendices contain the Glossary and supporting materials.

2 User Requirements Definition

This section defines the specific needs of the three primary user roles based on stakeholder input from students, tutors, and the administrator of an existing tutoring platform (Quickstudy).

2.1 Student

The system shall provide students (and guardians managing accounts for younger learners) with the following capabilities:

- Simple sign-up, login, and mobile-responsive interface.
- Easy search and selection of tutors by subject, experience, and reviews.
- Flexible scheduling, trial classes, and booking of live online sessions.
- Trial classes before committing to regular sessions.
- Transparent pricing with clear hourly rates, pay-per-class, or monthly package options.
- Basic interactive tools (chat, file sharing) during live sessions.
- Progress tracking, homework support, practice tests/quizzes, and post-session notes or summaries.
- Reminders for upcoming classes and quick support for missed classes or payment issues.

2.2 Tutor

The system shall enable tutors to:

- Complete registration with a short platform-based test and administrator video-call interview.
- Set fixed, non-negotiable hourly rates.
- Specify availability time slots (e.g., 8:00 pm–9:00 pm) that are mutually confirmed with students.
- Offer online classes only(also allows multiple students in the same time slot).
- Manage session links, conduct classes with interactive tools (chat with edit/delete messages, file sharing etc).

2.3 Administrator

The system shall allow administrators to:

- Manage tutor registration, conduct short platform tests, and perform video interviews.
- Verify tutor credentials and manage user accounts.
- Monitor platform activity, sessions, and complaints.
- Handle support requests related to classes, payments, or technical issues.

3 System Architecture

ShikkaLink shall follow a simple client-server architecture consisting of:

- A web-based frontend that provides role-specific interfaces for Students, Tutors, and Administrators.
- Backend services responsible for user authentication, scheduling, session management, and communication.
- A centralized database for storing user profiles, session data, and activity logs.
- External links for third-party video meeting tools.

All features and data shall be strictly controlled by user roles.

4 System Requirements Specification

4.1 Functional Requirements

The system shall meet the following functional requirements:

1. The system shall support simple user registration, minimal login, and role-based access for Students, Tutors, and Administrators.
2. The system shall allow tutors to complete a short platform-based test and administrator video-call interview.
3. The system shall let tutors specify teaching interests and upload a demo video in their profile.
4. The system shall allow tutors to set fixed, non-negotiable hourly rates that are clearly displayed.
5. The system shall enable mutual selection and confirmation of availability time slots.
6. The system shall support online classes only and allow multiple students in the same time slot.
7. The system shall allow searching and filtering of tutors by subject and experience.

8. The system shall use simple AI to recommend suitable tutors to students based on subject and availability.
9. The system shall enable booking of trial and regular sessions, including meeting links for online classes.
10. The system shall provide real-time chat and file sharing during sessions.
11. The system shall provide role-based dashboards for each user type.

4.2 Non-Functional Requirements

- **Usability**: The interface shall be simple, mobile-responsive, and have clear call-to-action buttons.
- **Performance**: The system shall respond quickly to searches, bookings, and chat messages.
- **Security and Privacy**: User data, payments, and login shall be secure with role-based access control.
- **Reliability**: The system shall work reliably during scheduled class times.
- **Accessibility**: The platform shall be accessible from any modern web browser on mobile or desktop.

5 System Models

This section presents models illustrating the behaviour and structure of ShikkaLink. It includes use-case diagrams, data models, and process flows for key activities such as tutor registration, session booking, and complaint handling. These models will be used to validate that all stakeholder requirements are addressed.

6 System Evolution

ShikkaLink is designed to evolve after the initial release. Future enhancements may include guardian-specific monitoring features, automated payment processing, group sessions, mobile applications, and advanced analytics, subject to stakeholder feedback and available resources.

7 Appendices

7.1 Glossary

- **ShikkaLink**: The proposed web-based online tutoring platform.
- **Student**: Primary users seeking tutoring services.
- **Tutor**: Qualified instructors offering paid online tutoring sessions.

- **Administrator**: Platform operator responsible for user verification, monitoring, and management.
- **Session**: A scheduled one-on-one online tutoring class conducted via external meeting links.
- **Tutor Verification**: Process involving a short platform-based test and optional administrator video-call interview.
- **Hourly Rate**: Fixed fee set by the tutor and displayed during booking.

7.2 Appendix A: Requirements Gathering Materials

All raw interview notes, questionnaires, and stakeholder suggestions (from tutors and students with experience on Dhikka.com, and the Quickstudy administrator) are documented in the file **ClientInterviewReport.pdf**.
The complete document can be accessed at:
https://drive.google.com/file/d/1hOm0CJ2dcPLTvu8eWr4KFAYBWgDY04oh/view
This PDF contains the full records of discussions, questions asked, and feedback received during the requirements elicitation phase.

7.3 Appendix B: Stakeholder Feedback Summary

Key findings from the collected feedback include:

- Strong need for tutor verification through short platform test and video interview.
- Requirement for flexible scheduling of online classes with mutual time-slot confirmation.
- Demand for transparent, non-negotiable hourly rates.
- Importance of simple mobile-friendly interface and basic real-time chat during sessions.